

harini

Generated May 20, 2026

Project Summary

****Project 'Harini' (SAILP) Summary****

****Project Overview:****

SAILP (Specialized AI Learning Platform) is an educational platform designed for developers and computer science students. The platform aims to provide a structured learning experience through a goal-setting and roadmap engine, interactive instruction, live visual technical demonstrations, and a real-time industry news feed – all contextually linked to each other.

****Purpose:****

The primary purpose of SAILP is to provide a personalized and dynamic learning experience for developers and computer science students. The platform seeks to bridge the gap between theoretical knowledge and practical application by offering interactive instruction, live visual demonstrations, and real-time industry news feed.

****Key Features:****

1. ****Goal Setting & Roadmap Engine:**** The platform features a goal-setting and roadmap engine that breaks down high-level learning goals into structured, persistent roadmaps.
2. ****Interactive Teaching Chatbot (Tutor):**** A conversational AI tutor that is contextually aware of the user's active roadmap item, providing explanations, analogies, and worked examples.
3. ****Visual Frontend:**** A live visual technical demonstration feature that showcases the application of theoretical concepts in real-world scenarios.
4. ****Real-time Industry News Feed:**** A feature that provides users with real-time industry news and updates, contextualized to their learning path.
5. ****Multi-Agent Architecture:**** The platform utilizes a multi-agent architecture to provide dynamic learning paths, interactive instruction, and live visual technical demonstrations.

****Technical Details:****

1. ****Development Framework:**** The platform is built using Python, with FastAPI as the web framework and LangGraph for graph wiring.
2. ****Database Integration:**** MySQL is used for user account and roadmap state management.
3. ****Chatbot Integration:**** The Tutor chatbot is integrated using Llama 3.1 70B via NVIDIA NIM, with token-by-token response streaming via SSE.
4. ****Spaced Repetition Scheduling:**** The platform utilizes spaced repetition scheduling for concept reinforcement and knowledge gap detection.

****Current Status:****

The project is currently in the development phase, with the following milestones achieved:

1. **v1.0 (Core MVP):** Goal submission and Researcher Agent roadmap generation, persistent roadmap sidebar with checkbox progress tracking, Tutor chatbot with SSE streaming and session memory, and MySQL integration for user accounts and roadmap state.
2. **v1.1 (Visual Layer):** Developer Agent integration – Topic of the Day generation, and CSV data integration.

Important Findings:

1. **Multi-Agent Architecture:** The multi-agent architecture enables the platform to provide dynamic learning paths, interactive instruction, and live visual technical demonstrations.
2. **Spaced Repetition Scheduling:** The spaced repetition scheduling feature helps reinforce concepts and detect knowledge gaps, improving the overall learning experience.
3. **Real-time Industry News Feed:** The real-time industry news feed feature provides users with contextualized industry updates, enhancing their learning experience.

Conclusion:

SAILP is a cutting-edge educational platform designed to provide a structured learning experience for developers and computer science students. The platform's key features, including the goal-setting and roadmap engine, interactive teaching chatbot, and real-time industry news feed, make it an ideal solution for students seeking a personalized and dynamic learning experience. With its multi-agent architecture and spaced repetition scheduling feature, SAILP has the potential to revolutionize the way students learn and absorb knowledge.

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SAILP (Specialized AI Learning Platform) is an educational platform designed for developers and computer science students. It utilizes a multi-agent architecture to provide dynamic learning paths, interactive instruction, live visual technical demonstrations, and a real-time industry news feed.

Key Features:

1. **Goal Setting & Roadmap Engine (The Architect):** Breaks down high-level learning goals into structured, persistent roadmaps.
2. **Interactive Teaching Chatbot (The Tutor):** A conversational AI tutor that provides contextually aware responses to user queries, scoped to the current learning phase.
3. **Visual Frontend Demonstrator (The Developer Agent):** Generates live, rendered React or

HTML/CSS components to demonstrate abstract concepts.

4. **Real-Time Contextual News Feed (Industry Updates)**: Displays filtered news relevant to the user's current learning phase.

Technical Details:

1. **Agent 1 - Researcher Agent**: Breaks down high-level learning goals into structured roadmaps.
2. **Agent 2 - Tutor (Interactive)**: Utilizes Llama 3.1 70B via NVIDIA NIM to provide interactive teaching.
3. **MySQL Integration**: Stores user accounts and roadmap state.
4. **FastAPI + LangGraph**: Provides basic graph wiring.

Key Findings:

1. **Dynamic Roadmap Reshaping**: Roadmaps are dynamically reshaped based on user session behavior.
2. **Spaced Repetition Scheduling**: Concepts are reinforced through spaced repetition scheduling.
3. **Cross-Session Memory**: Long-term learner profiles are maintained across sessions.
4. **Performance Tracking**: Knowledge gap detection and performance tracking are integrated.

Project Roadmap:

1. **v1.0 (Core MVP)**: Goal submission, Researcher Agent roadmap generation, and basic features.
2. **v1.1 (Visual Layer)**: Developer Agent integration, Topic of the Day generation, and visual frontend enhancements.

Project Structure:

1. **SAILP v1.0**: End of documentation, May 2026.
2. **Agent 1 - Researcher Agent**: Breaks down high-level learning goals into structured roadmaps.
3. **Agent 2 - Tutor (Interactive)**: Utilizes Llama 3.1 70B via NVIDIA NIM to provide interactive teaching.

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Key Features:

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Key Features:

1. **Goal Setting & Roadmap Engine:** The Researcher Agent breaks down high-level learning goals into structured, persistent roadmaps.
2. **Interactive Teaching Chatbot:** The Tutor is a conversational AI that provides contextually aware responses to the user's active roadmap item.
3. **Visual Frontend Demonstrations:** The platform features a visual frontend that showcases the capabilities of the platform.

Technical Requirements:

1. **Core MVP (v1.0):** The initial release will include:
 - * Goal submission and Researcher Agent roadmap generation
 - * Persistent roadmap sidebar with checkbox progress tracking
 - * Tutor chatbot with SSE streaming and session memory
 - * MySQL integration for user accounts and roadmap state
 - * Basic FastAPI + LangGraph graph wiring
2. **Visual Layer (v1.1):** The next release will include:
 - * Developer Agent integration for Topic of the Day generation

Key Findings:

1. The platform requires a robust backend infrastructure to support the goal-setting and roadmap engine, interactive teaching chatbot, and visual frontend demonstrations.
2. The use of FastAPI and LangGraph will enable efficient graph wiring and data processing.
3. MySQL integration will provide a reliable and scalable storage solution for user accounts and roadmap state.

****Next Steps:****

1. Develop the Core MVP (v1.0) to establish a solid foundation for the platform.
2. Integrate the Developer Agent for Topic of the Day generation in the Visual Layer (v1.1) release.
3. Continuously iterate and refine the platform to ensure a seamless user experience.

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